Name & Surname: St number:
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Please use this document to answer the questions.

3 Theory component (30 marks)

3.1 Model scenario (3 marks)

Give the scenario of your model. How is this model useful in the context of cybersecurity?

3.2 Model architecture (10 marks)

Explain all relevant details of the model you used. For example, if you implemented a Convolutional Neural Network (CNN), specify the types of layers, the activation functions used, and the overall model architecture. Also explain why the chosen model is well suited to the data it was trained on.

3.3 Model result analysis (10 marks)

Explain the results of your model using the metrics and graphs derived from the model's training. Is this model effective in the problem it is trying to resolve?

3.4 Real-world use of models (7 marks)

Evaluate the real-world use of models within the field of cybersecurity, focusing on how these models are applied to tasks such as intrusion detection, threat classification, anomaly detection, and risk assessment. Discuss their effectiveness in practical deployments, including strengths, limitations, and common challenges such as data quality, concept drift, and false positives. Use academic references.